1. **Problem 1:**
   i. **Different results, I would like to explain:**

In the main method:

```
Person p1 = new PersonWithJob("Joe", 30000);
Person p2 = new Person("Joe");
```

Two comparisons are made:

```
p1.equals(p2);
p2.equals(p1);
```

Comparation 1:

p1.equals(p2) => false. Because: p1 is an instance of PersonWithJob. So, the equals method in PersonWithJob is executed. This method checks:

```
if(!(aPerson instanceof PersonWithJob)) return false;
```

p2 is only a Person, not a PersonWithJob **=>** condition fails **=>** Result: false

Comparation 2:

p2.equals(p1) → true. Because: p2 is an instance of Person. So, the equals method in Person is executed. This method checks:

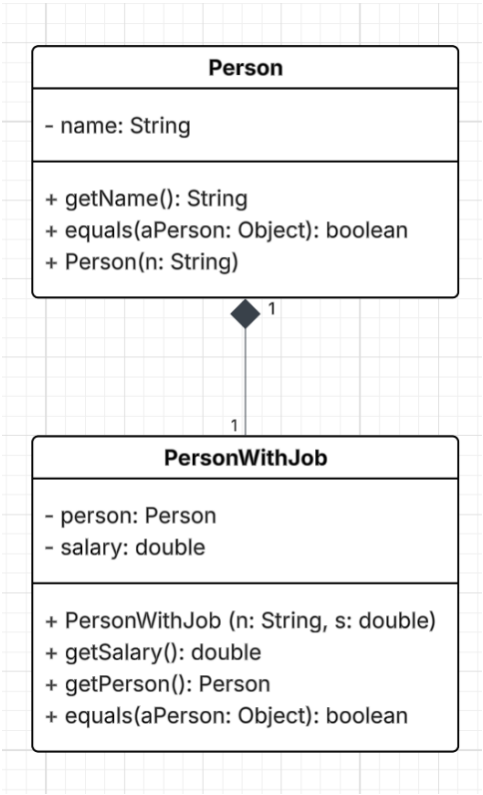
```
if(!(aPerson instanceof Person)) return false;
```

p1 is a PersonWithJob, which is a subclass of Person => condition passes. Then it compares names: both are "Joe" => equal => Result: true

ii. A solution by replacing inheritance with composition:

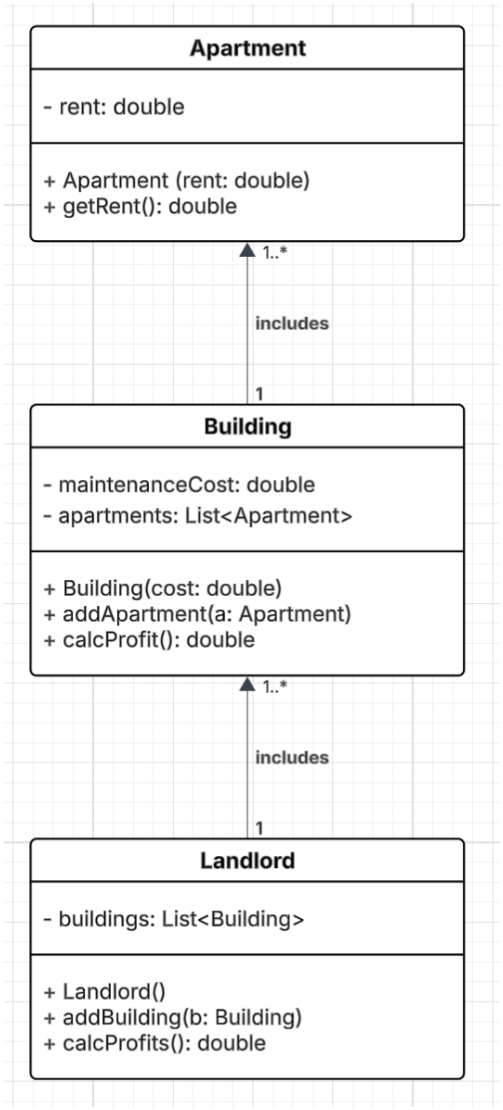
i. **New code**: https://github.com/miunthphuong/MPP/tree/main/Assigment/Lab%203/Lab3%20-%20code/code%20for%20prob1/lesson3/labs/prob1

ii. New Class Diagram:



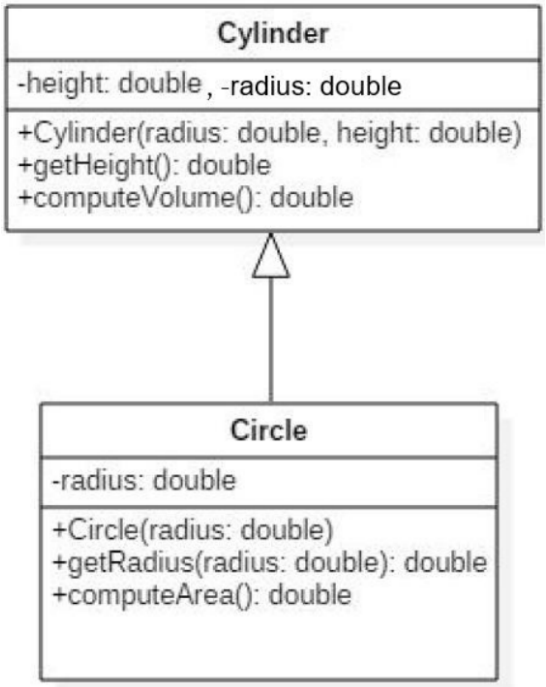
2. **Problem 2:**

   i. **Solution with class diagram:**



   ii. **Source code:**
   https://github.com/miunthphuong/MPP/tree/main/Assigment/Lab%203/Lab3%20-%20code/code%20for%20prob2

3. **Problem 3:** The class diagram

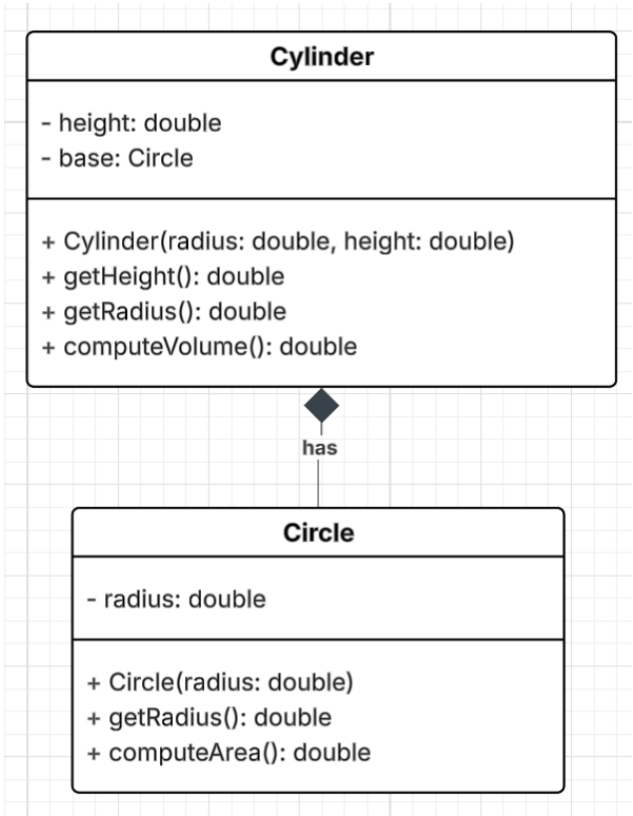


i. **This inheritance in above class diagram does not make sense**. Because Inheritance expresses an "is-a" relationship. So the diagram is saying: *Cylinder is a Circle*. That's incorrect in object-oriented design. The real relationship: *A cylinder is NOT a circle*. Instead: A cylinder has a circular base. => This is a composition (has-a) relationship, not inheritance.

Source code:
https://github.com/miunthphuong/MPP/tree/main/Assigment/Lab%203/Lab3%20-%20code/code%20for%20prob3/prob3-A/src

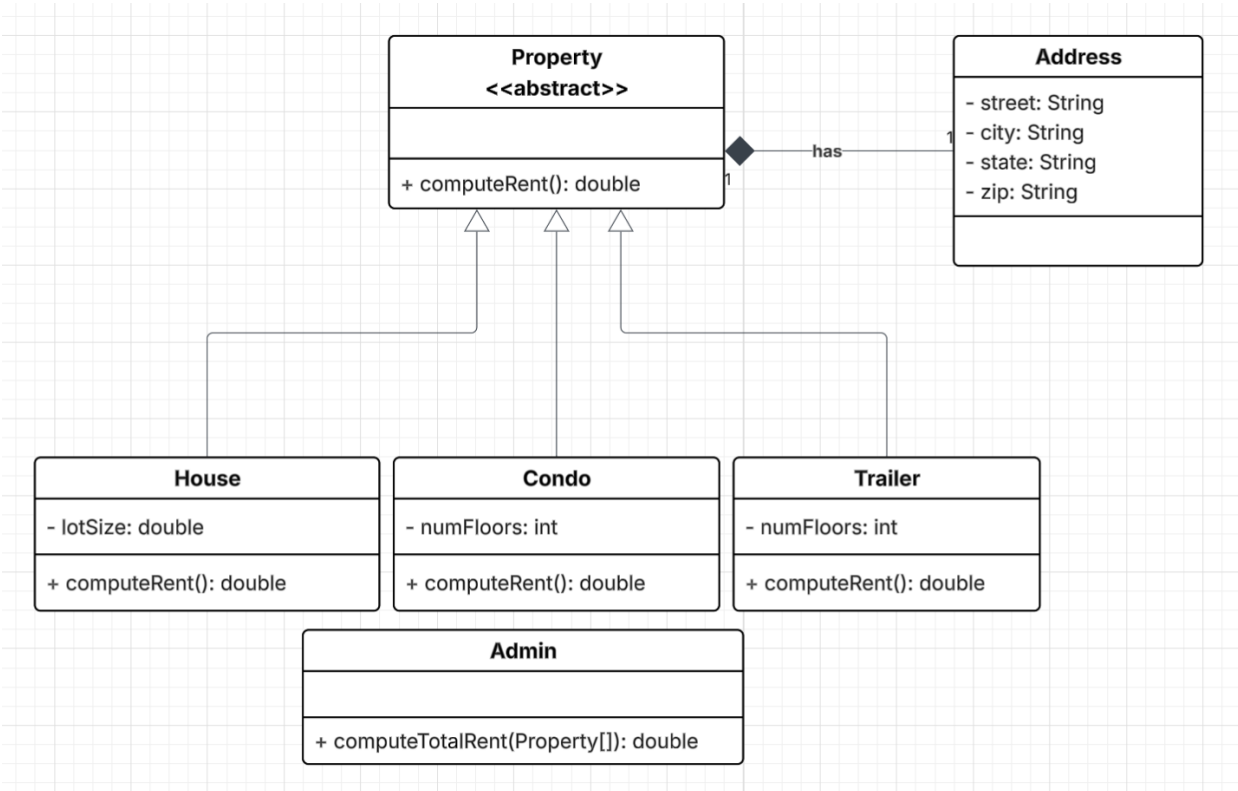
ii. Using composition

Class diagram:



Source code:
https://github.com/miunthphuong/MPP/tree/main/Assigment/Lab%203/Lab3%20-%20code/code%20for%20prob3/prob3-B/src

4. **Problem 4:**
   i. Class diagram:



   ii. Source code: https://github.com/miunthphuong/MPP/tree/main/Assigment/Lab%203/Lab3%20-%20code/code%20for%20prob4